

BLACKPOOL, United Kingdom

WMO: 033180

Lat: 53.774N

Lon: 3.039W

Elev: 10

StdP: 101.20

Time Zone: 0.00 (EUW)

Period: 10-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-3.3	-1.8	-5.9	2.3	-0.8	-4.5	2.6	0.0	28.3	7.1	26.2	7.0	2.6	100	0.715

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	6.1	23.7	17.2	21.5	16.6	20.1	16.0	18.3	21.6	17.5	20.2	16.8	19.1	5.5	100

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.2	12.3	19.4	16.5	11.7	18.6	15.8	11.2	18.0	51.9	21.7	49.3	20.3	47.3	19.3	23.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 22.8	19.1	14.0	DB	-4.8	28.5	1.0	2.1	-5.5	30.0	-6.0	31.3	-6.5	32.5	-7.2	34.0	(4)
(5)			WB	-5.2	20.9	1.1	1.5	-6.0	21.9	-6.6	22.8	-7.2	23.6	-8.0	24.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	10.2	5.0	5.1	6.2	8.7	11.6	14.3	16.3	16.0	14.0	11.2	7.4	5.9	(6)	
	DBStd	4.86	2.79	2.62	2.63	2.71	2.82	2.19	2.01	1.65	2.11	2.59	3.09	3.40	(7)	
	HDD10.0	716	157	138	121	57	13	1	0	0	1	17	86	127	(8)	
	HDD18.3	2993	415	371	377	290	211	122	70	75	131	221	327	384	(9)	
	CDD10.0	776	0	0	2	17	62	130	195	185	121	54	8	1	(10)	
	CDD18.3	13	0	0	0	0	1	2	7	2	1	0	0	0	(11)	
	CDH23.3	82	0	0	0	0	11	21	43	4	2	1	0	0	(12)	
	CDH26.7	12	0	0	0	0	1	3	7	1	0	0	0	0	(13)	
(14) Wind	WSAvg	6.1	7.0	6.7	5.8	5.4	5.7	5.2	5.2	6.0	5.8	6.1	6.5	7.6	(14)	
(15) Precipitation	PrecAvg	957	88	71	58	54	60	62	67	86	89	108	105	104	(15)	
	PrecMax	1252	151	150	133	91	122	146	142	189	175	244	195	252	(16)	
	PrecMin	729	8	13	21	16	14	27	26	14	9	28	54	27	(17)	
	PrecStd	142	36	34	30	22	27	30	24	36	38	44	39	51	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	11.5	13.0	16.0	20.8	25.0	26.0	27.7	23.4	22.5	18.9	14.5	13.0	(19)	
		MCWB	10.3	9.5	11.0	14.1	16.6	17.3	18.9	17.6	17.3	15.6	13.4	11.7	(20)	
	2%	DB	10.0	10.4	12.7	17.7	21.1	22.0	24.0	21.1	20.1	17.0	13.6	11.6	(21)	
		MCWB	9.1	8.9	10.3	11.7	15.5	16.3	17.6	16.9	16.2	14.7	12.4	10.4	(22)	
	5%	DB	9.1	9.5	11.0	14.8	18.0	19.7	21.5	19.8	18.7	15.9	12.6	10.8	(23)	
		MCWB	8.2	8.4	8.9	10.7	13.5	15.4	17.1	16.4	15.4	14.2	11.5	9.8	(24)	
	10%	DB	8.5	8.6	9.8	12.7	15.8	18.2	19.9	18.7	17.4	14.9	11.7	10.0	(25)	
		MCWB	7.6	7.6	8.2	10.1	12.6	14.7	16.4	15.7	14.8	13.2	10.5	9.0	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.5	10.1	12.3	14.5	17.6	18.9	20.4	18.8	18.6	16.4	13.7	11.9	(27)	
		MCDB	11.1	11.2	14.5	20.0	23.6	23.9	26.4	21.2	21.6	18.0	14.3	12.7	(28)	
	2%	WB	9.2	9.2	10.6	12.7	15.8	16.9	18.6	17.8	17.1	15.4	12.7	10.8	(29)	
		MCDB	9.7	10.2	12.5	16.4	20.2	20.4	22.1	19.9	19.0	16.7	13.3	11.4	(30)	
	5%	WB	8.5	8.6	9.4	11.5	14.4	16.0	17.7	17.1	16.1	14.6	11.7	10.0	(31)	
		MCDB	9.1	9.4	10.7	13.8	17.2	19.1	20.7	19.0	17.7	15.8	12.4	10.6	(32)	
	10%	WB	7.9	7.9	8.6	10.4	13.1	15.2	16.9	16.5	15.3	13.5	10.7	9.2	(33)	
		MCDB	8.5	8.6	9.7	12.6	15.3	17.8	19.3	18.3	16.9	14.7	11.6	10.0	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	4.0	4.6	6.0	6.7	6.8	6.6	6.1	5.4	6.0	5.5	4.8	4.5	(35)	
		MCDBR	3.7	4.9	7.8	10.6	10.7	10.1	9.8	7.3	8.4	6.4	4.4	4.8	(36)	
	5% WB	MCWBR	3.8	3.9	5.6	6.2	6.1	5.7	5.2	4.4	5.1	4.8	4.2	4.7	(37)	
		MCWBR	3.9	4.6	6.8	8.6	9.7	8.4	7.8	6.3	6.2	5.5	4.1	4.6	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.330	0.341	0.369	0.390	0.394	0.392	0.393	0.379	0.366	0.350	0.327	0.317	(40)	
		taud	2.307	2.375	2.316	2.283	2.318	2.353	2.355	2.407	2.443	2.442	2.376	2.282	(41)	
	Ebn at Noon		620	752	805	837	853	858	849	841	805	733	625	552	(42)	
		Edh at Noon	60	77	100	117	119	116	114	102	88	71	56	51	(43)	
(44) All-Sky Solar Radiation	RadAvg		0.61	1.29	2.44	3.80	4.87	5.05	4.81	3.93	2.79	1.53	0.75	0.46	(44)	
	RadStd		0.07	0.14	0.20	0.34	0.36	0.47	0.47	0.26	0.21	0.14	0.08	0.04	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (67 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page